

NUMERICAL FINANCE PROJECT

PRICING OF BERMUDAN BASKET OPTIONS

Lecturer : Kaiza AMOUH

Deadline : May 3rd 2026

SCOPE : The goal of this project is to price a Bermudan Basket Option using Monte Carlo simulation, and combine several variance reduction methods in order to reach a minimal simulation variance. The underlying diffusion is assumed to be a multi-dimensional Black-Scholes diffusion

The project can be implemented either in C++ or Python

- A main program must define and control the parameters to use, then call different Classes to perform the simulation
- The project's architecture is free, and its design will particularly be appreciated.
- The program must be fully commented and parameters must be checked for consistency.

In addition to your program, you must also produce a User Guide PDF file that :

- Explains the modifiable variables in the main function
- Illustrates your results for different sets of inputs through graphs and tables
- Details your analysis and comments

DELIVERY : This project must be sent by **May 3rd 2026** at **23:59** to the following address : kaiza.amouh@gmail.com

The sent email **MUST** contain the complete names of all team members. You can either attach your program and user guide, or send a **WeTransfer** link if your files are too heavy.

1. Given n correlated assets $S^1, S^2, \dots, S^n$, and some (possibly negative) weights $\alpha_1, \alpha_2, \dots, \alpha_n$, a *European Basket Call* option pays the following payoff only at maturity T .

$$\left(\sum_{i=1}^n \alpha_i S_T^i - K \right)^+ \quad (1)$$

- (a) Perform a Monte Carlo Simulation using basic Pseudo-Random numbers, without implementing any variance reduction method.
 - (b) Show the gain in variance and the gain in required number of simulations to enter a given confidence interval. You should cumulatively include the following variance reduction methods:
 - (i) Quasi-Random numbers
 - (ii) Static Control Variate
 - (iii) Antithetic Random variables
2. A *Bermudan* option with exercise dates $t_0 = 0, t_1, \dots, t_N = T$ pays upon the (random) exercise date τ , the amount

$$\left(\sum_{i=1}^n \alpha_i S_\tau^i - K \right)^+ \quad (2)$$

- (a) Using basic Pseudo-Random numbers without any variance reduction, implement the *Longstaff-Schwarz* algorithm to price this option.
- (b) Combining all the above-mentioned variance reduction methods, show the gain in variance and the gain in required number of simulations to enter a given confidence interval

Simulation of a correlated Gaussian Vector

Let $X \sim \mathcal{N}(\mu, \Sigma)$ be a *Gaussian Vector*. If Σ is a diagonal matrix, one can independently simulate each $\mathcal{N}(0, 1)$ component of the vector.

If Σ is not diagonal, components X_i must be simulated as linear combinations of $Y_i \sim \mathcal{N}(0, 1)$

$$X = \mu + BY, \text{ where } B \text{ is a square matrix satisfying } C = BB^t \quad (3)$$

- If Σ is invertible, its Cholesky decomposition generates a lower triangular matrix B
- If not, since Σ is always positive semidefinite, one can use the diagonalization process and find

$$\Sigma = ODO^t \text{ where } O \text{ is an Orthogonal matrix and } D \text{ Diagonal} \quad (4)$$

One can then choose $B = OD^{\frac{1}{2}}$